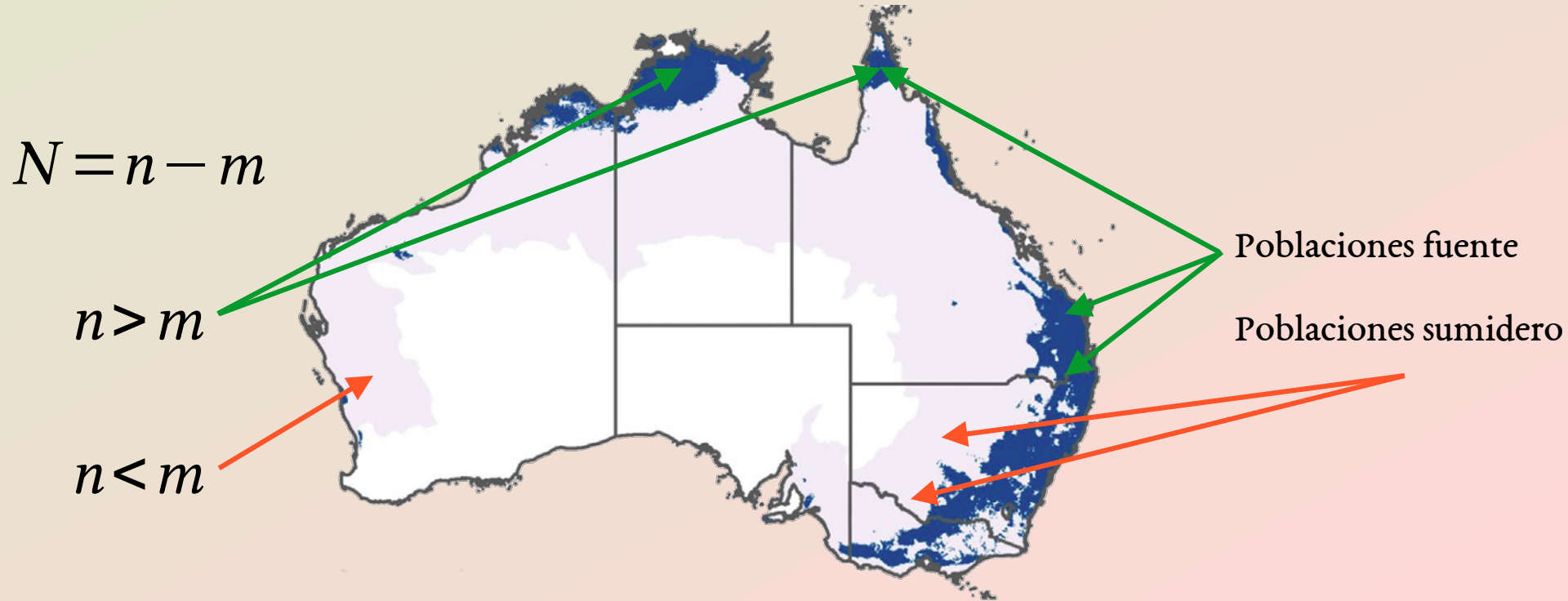


Perspectiva demográfica de los nichos ecológicos

Ecología Teórica II

¿Cómo se relacionan los nichos ecológicos con los
parámetros demográficos?

A nivel distribucional...



Tamaño de población, balance

$$N = n - m$$

$$n > m$$

$$n < m$$

¿Cuál(es) afectado por clima?

$$n(T^{\circ}, P, R, \dots) = \beta_1 T^{\circ} + \beta_2 P + \beta_3 R + \dots$$

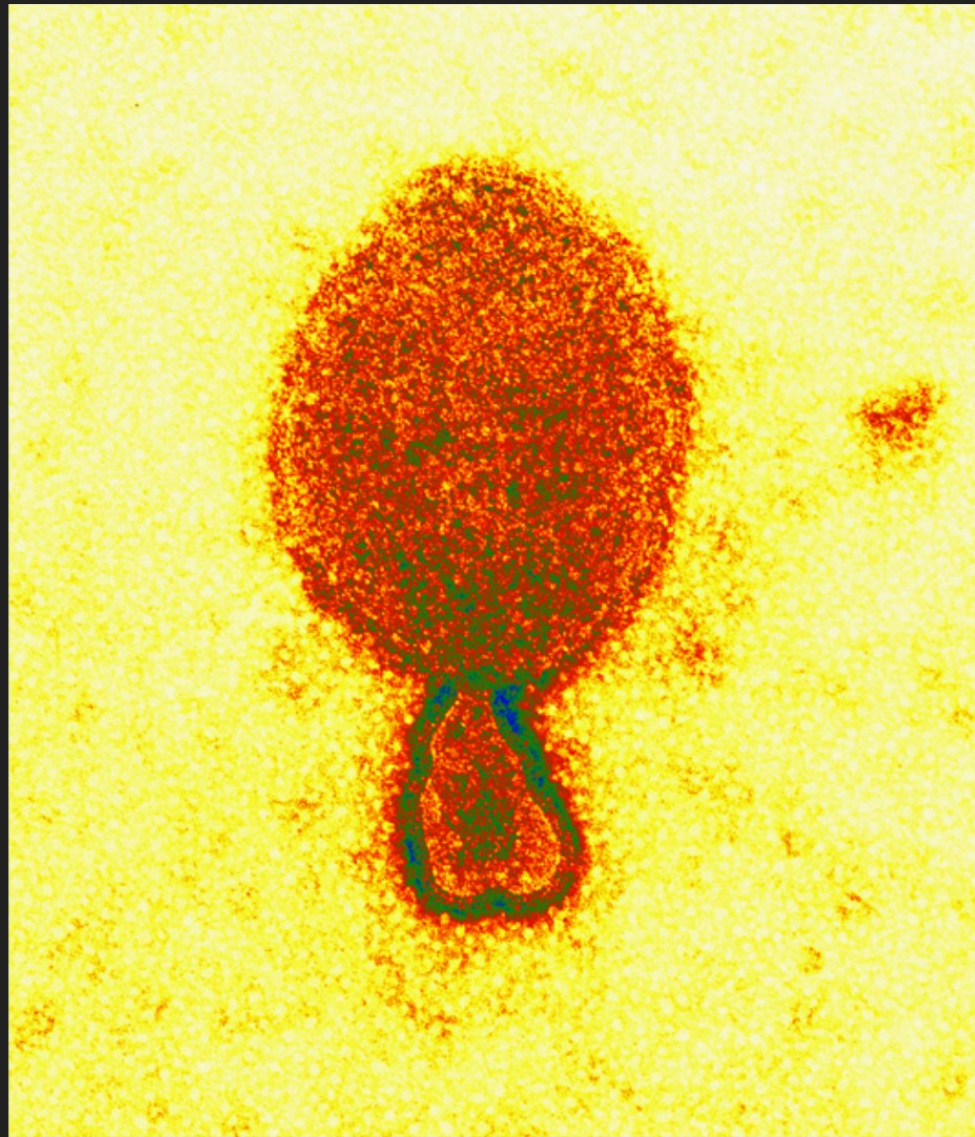
$$m(T^{\circ}, P, R, \dots) = \beta_1 T^{\circ} + \beta_2 P + \beta_3 R + \dots$$

Cantidad neta de nacimientos ó muertes son funciones de variables climáticas...

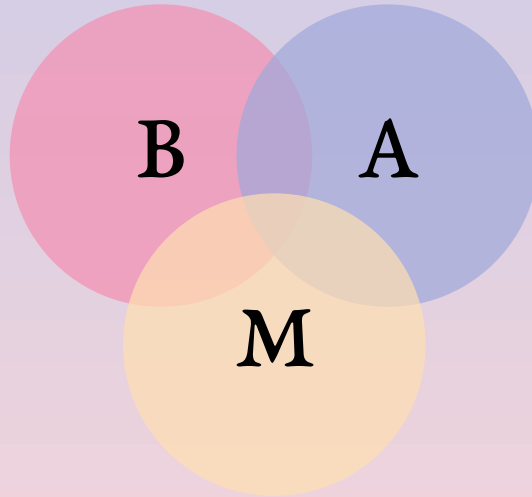
“Sólo es cuestión de obtener los datos ...”



Pues, nooooo, mejor trabajaremos con

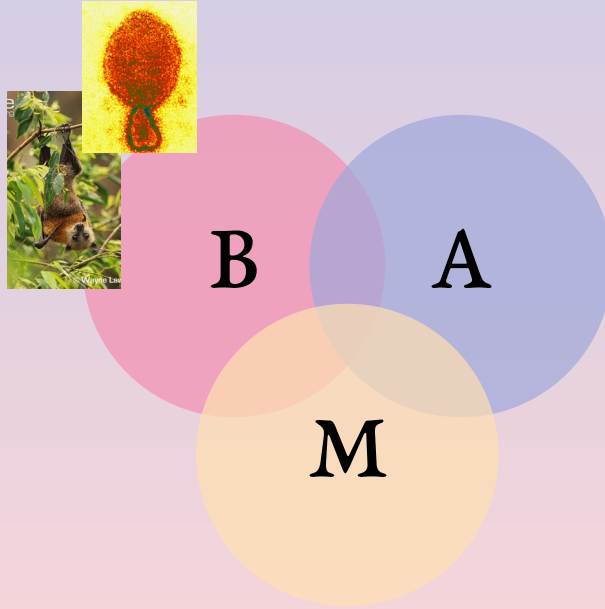


Nichos de organismos parásitos

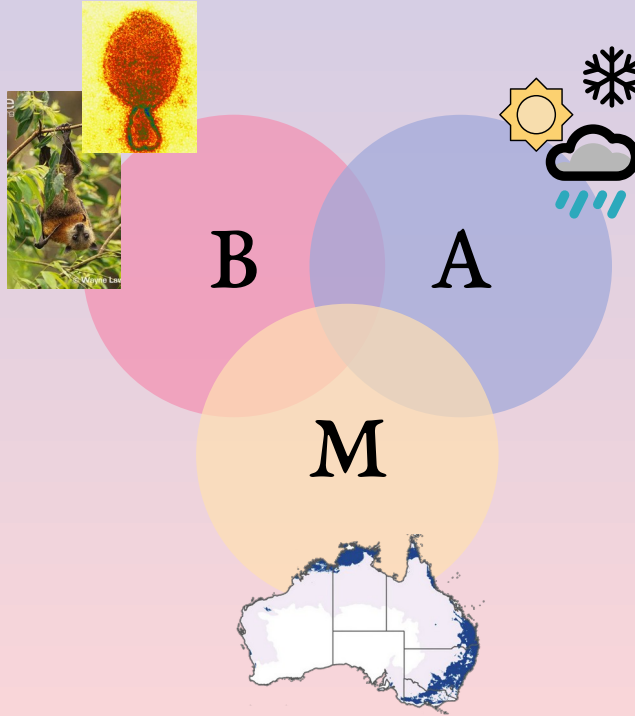


¿Cómo está conformado su diagrama?

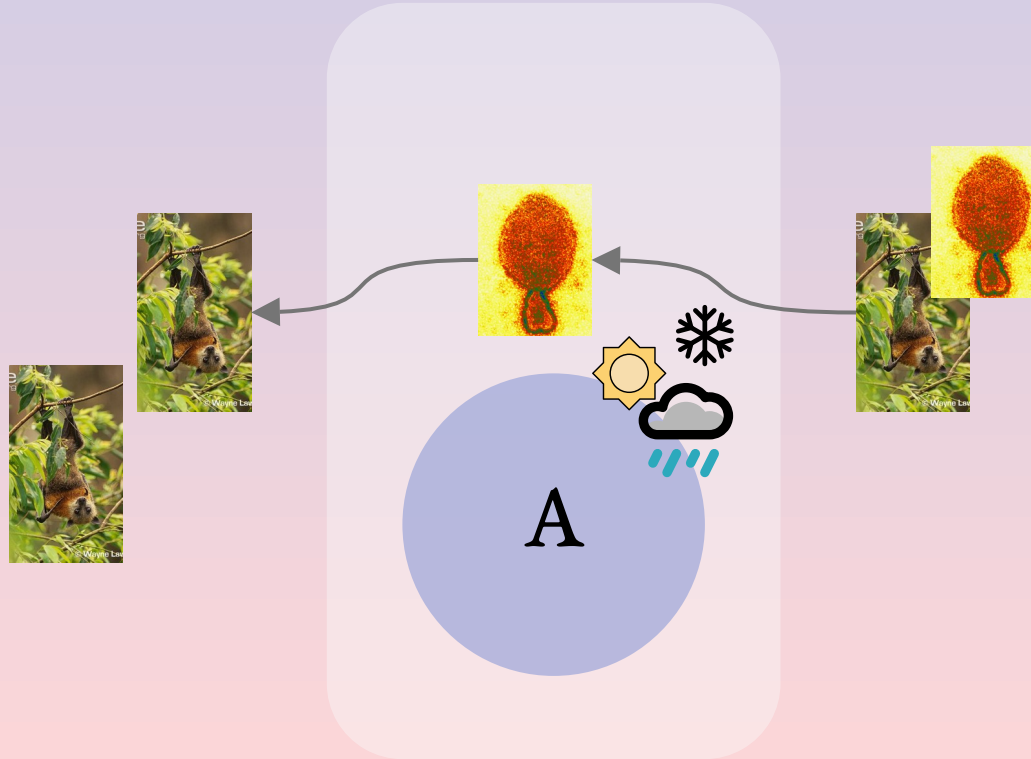
Nichos de organismos parásitos



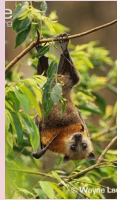
Nichos de organismos parásitos

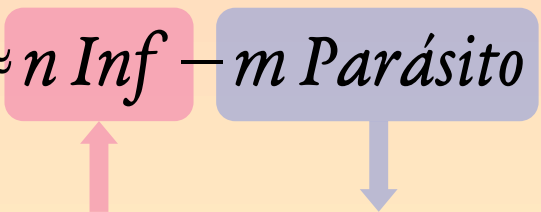


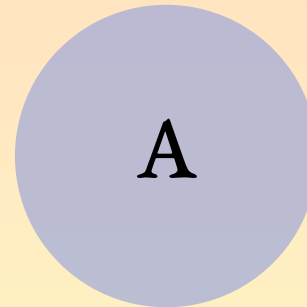
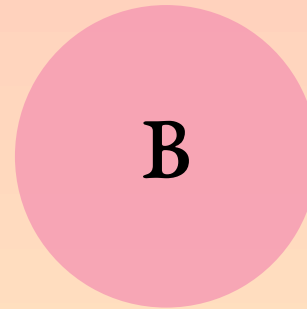
Nichos de organismos parásitos



Nichos de organismos parásitos

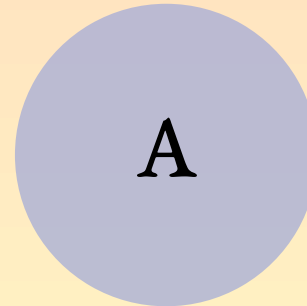


$$Parásito \approx n \text{ Inf} - m \text{ Parásito}$$




Experimentalmente
es más fácil estimar
 m como función del
ambiente

$$Parásito \approx n Inf - m Parásito$$



Por simplicidad, nos enfocaremos en:

$$m(T^{\circ}, P, R, \dots) = \quad {}_1T^{\circ} + \quad {}_2P + \quad {}_3R + \dots$$

Modelar el proceso de *muerte* como función de
condiciones físicas

Recordatorio

$$\frac{dS}{dt} = -\mu S$$

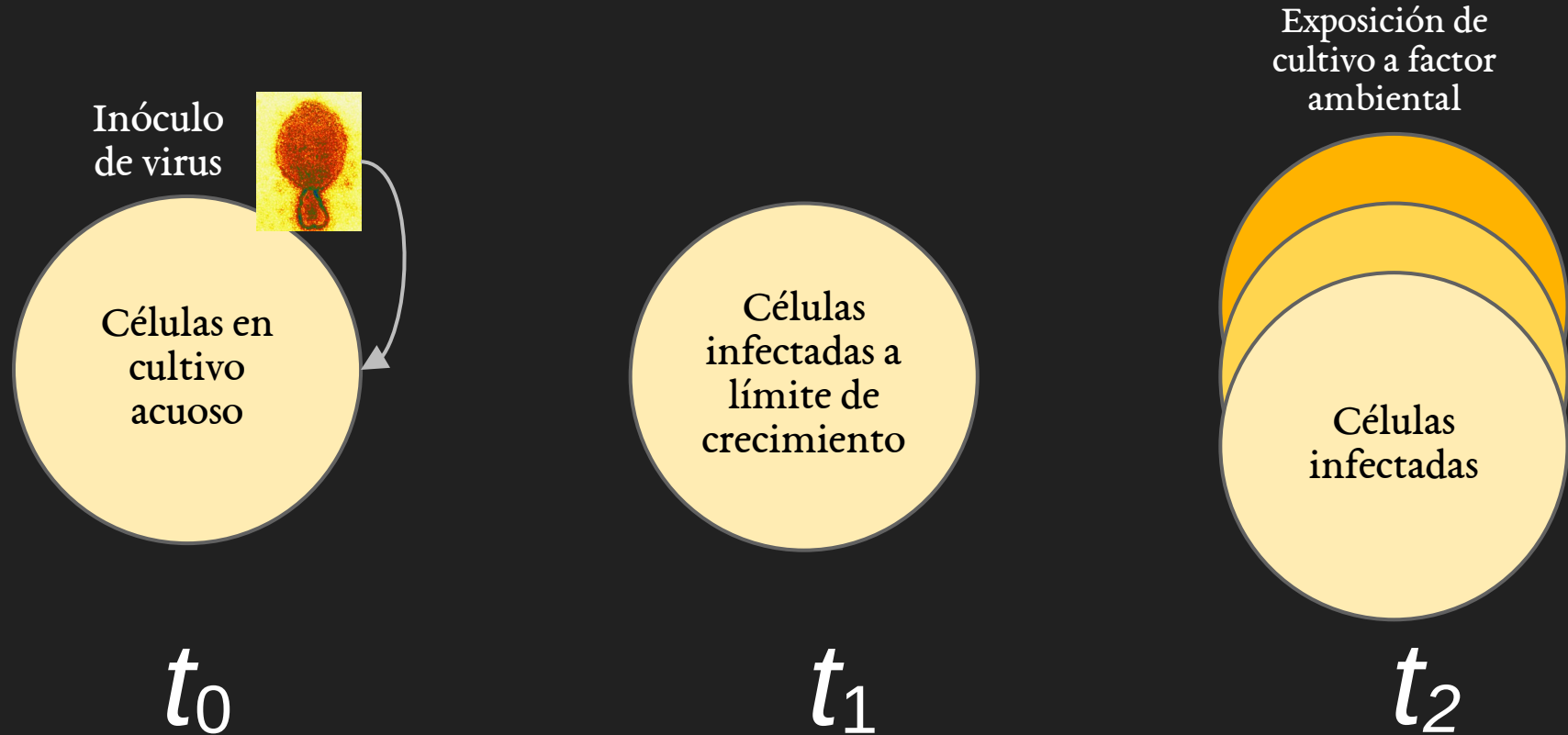
$$S(t) = S_0 e^{-\mu t}$$

Ignoraremos el caso donde $r > 0$, o sea que nuestros virus **no** se reproducen

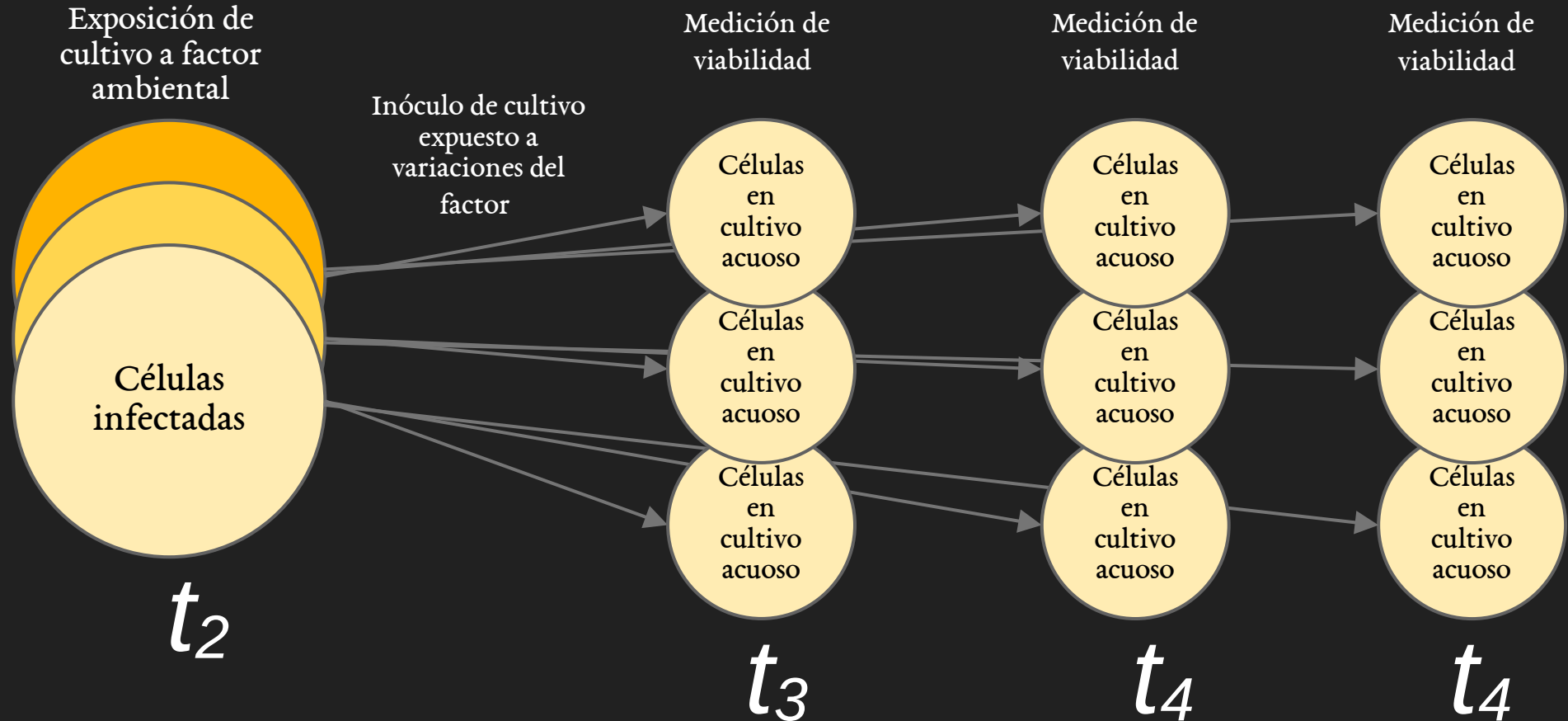
¿Qué requerimos para generar
 $\mu(T^\circ, \dots)$?

El diseño experimental

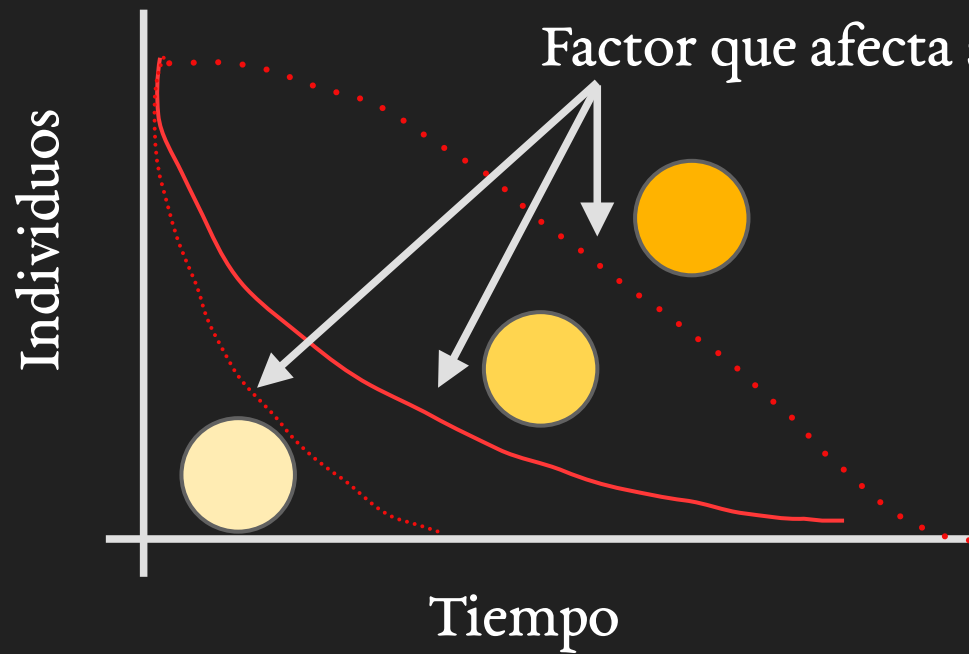
Cultivo de virus y medición de viabilidad



Cultivo de virus y medición de viabilidad

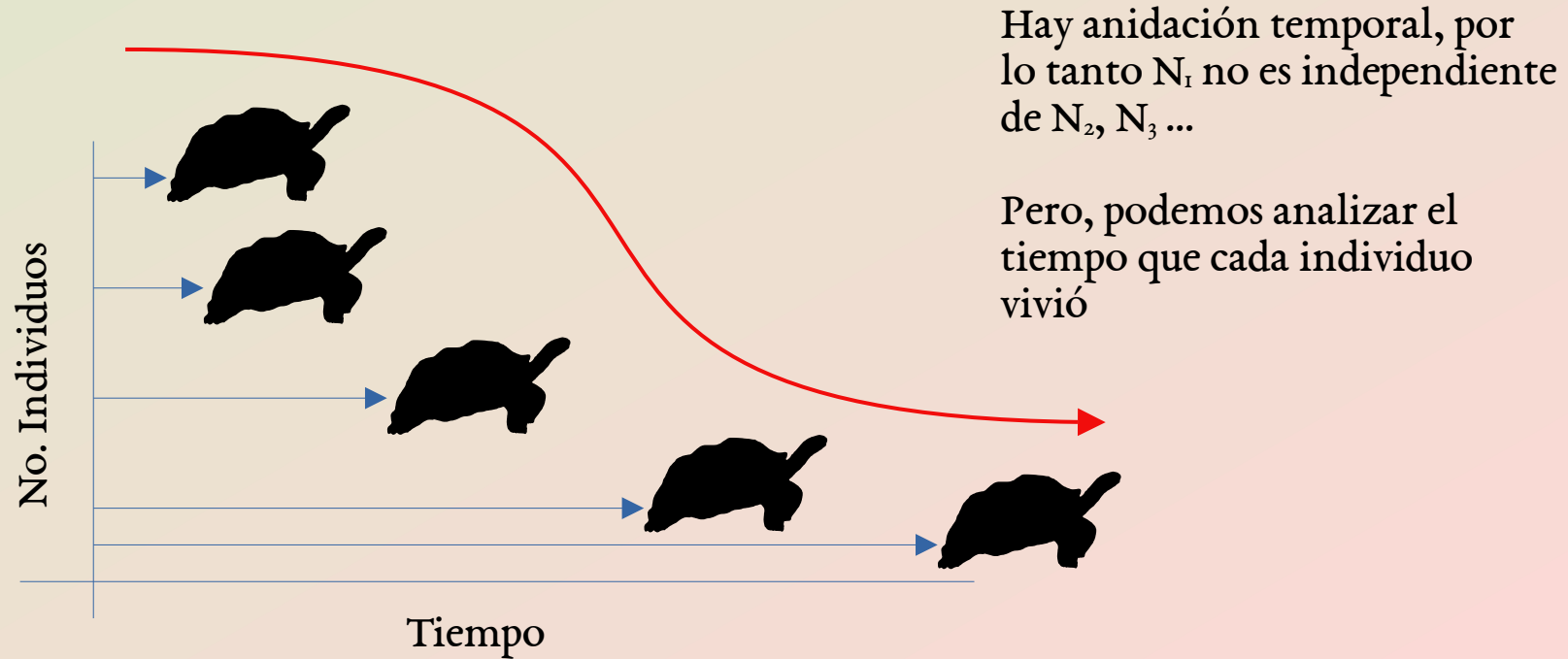


Paréntesis: Análisis de supervivencia



Ajustar curva para
cada nivel del
factor?

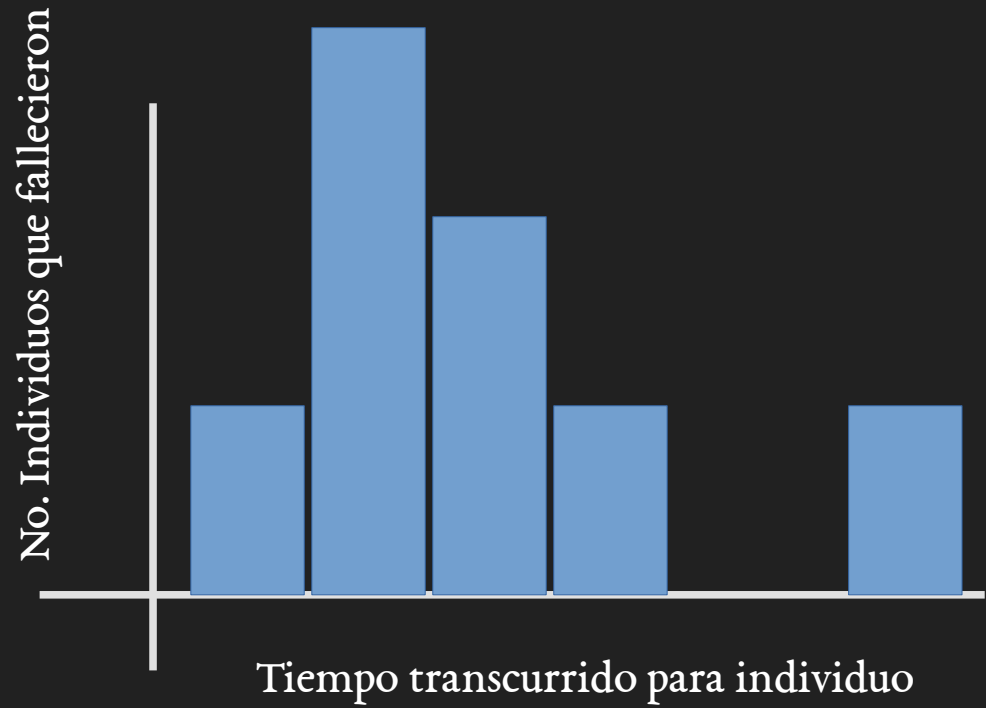
El problema



Análisis de supervivencia:

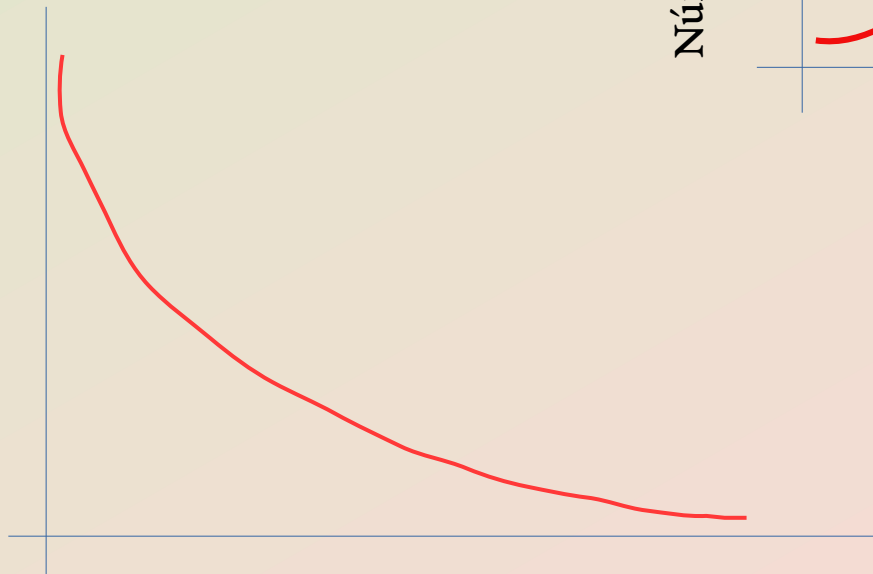
Estadísticas del tiempo transcurrido hasta que ocurre
evento

Individuo	Tiempo
1	0.5
2	1
3	1
4	1
5	1.5
6	1.5
7	2
8	4



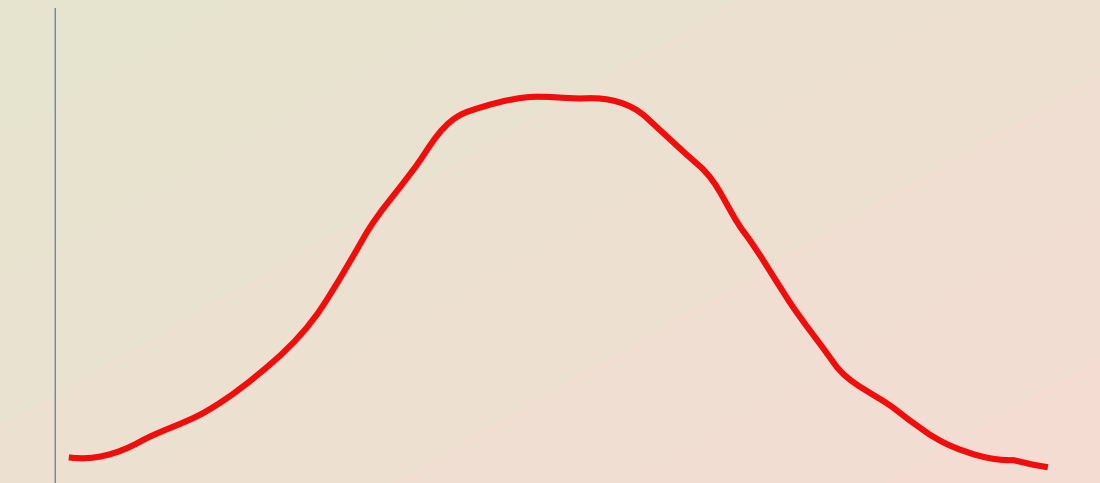
$$s(t) = \lambda e^{-\lambda} t$$

INDIVIDUOS



Tiempo

Número de individuos



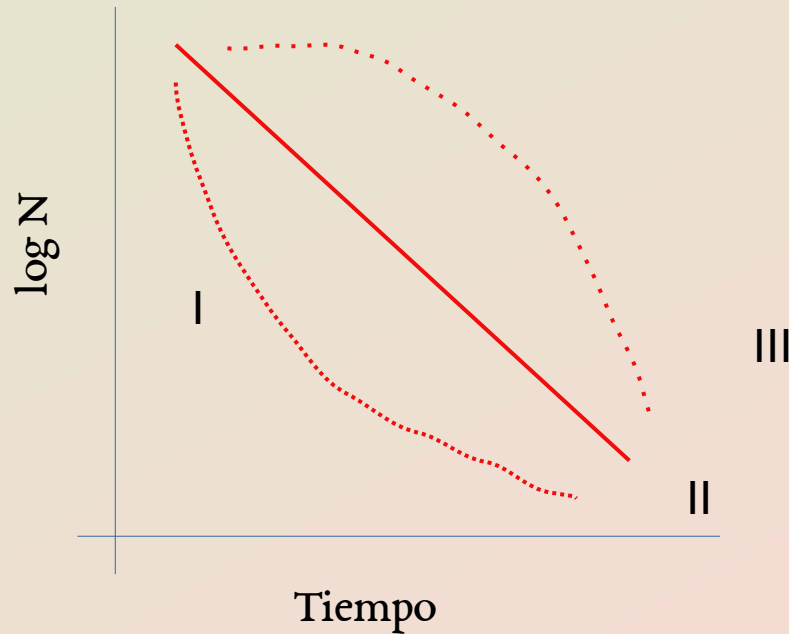
Tiempo transcurrido hasta evento (muerte, p. ej.)

$$S(t) = e^{-\lambda} t$$

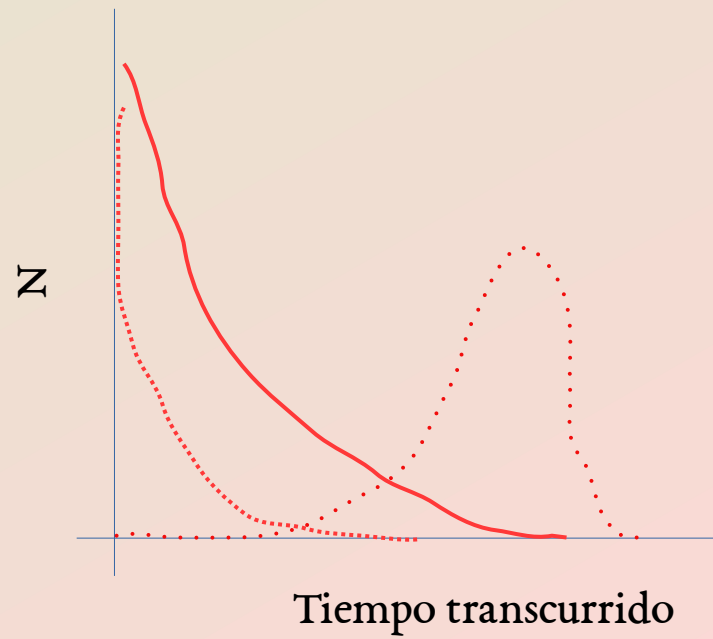
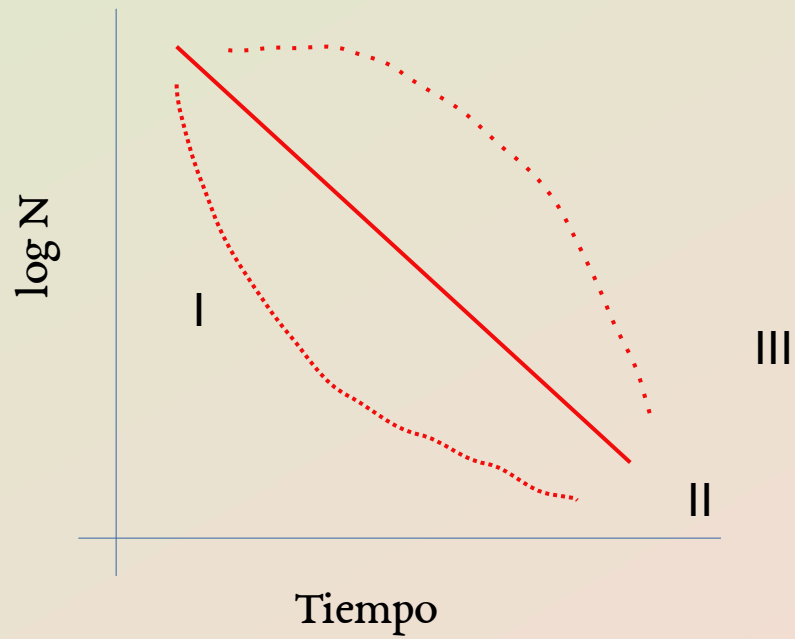
$$S(t) = \int s(t) dt$$

Gráficos sólo son esquemáticos

Tipos de curva de supervivencia

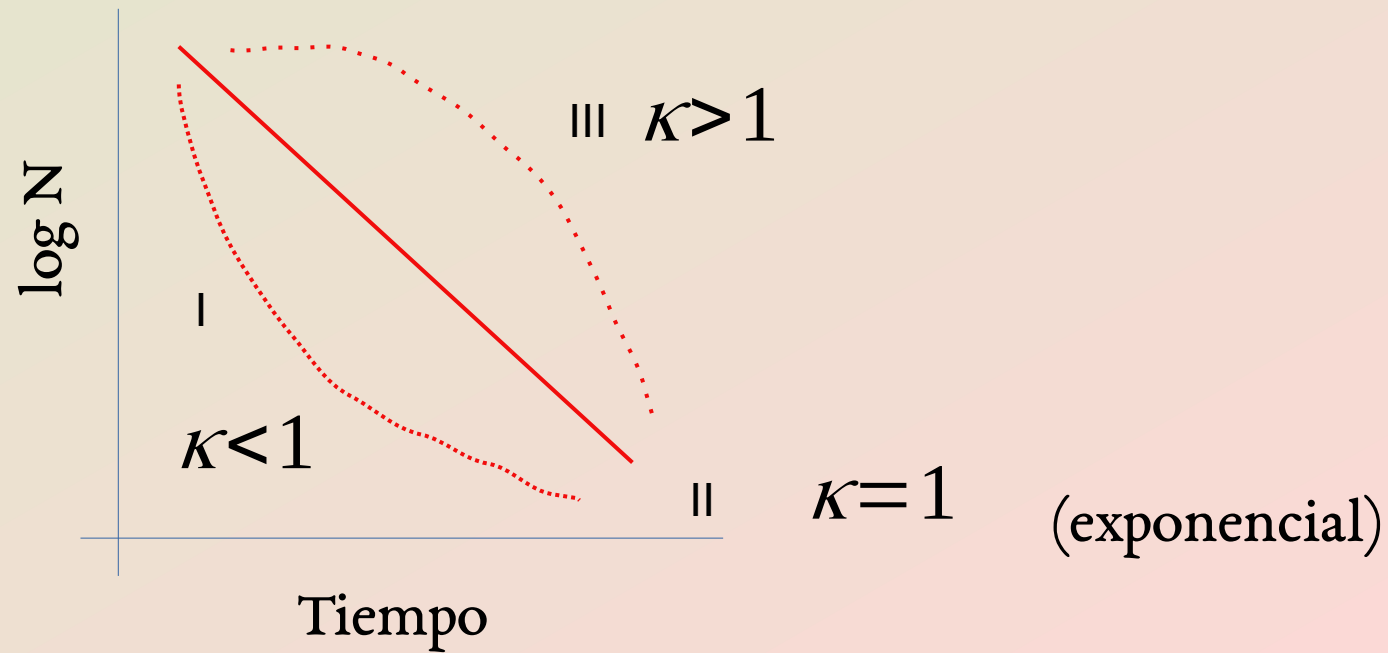


- I $\rightarrow$ Supervivencia aumenta con tiempo
- II $\rightarrow$ Supervivencia es constante
- III $\rightarrow$ Supervivencia disminuye



Distribuciones de tiempos

Distribución	Función de densidad	Función de supervivencia
Exponencial	$\rho e^{-\rho t}$	$e^{-\rho t}$
Weibull	$\kappa \rho (\rho t)^{\kappa-1} e^{-(\rho t)^\kappa}$	$e^{-(\rho t)^\kappa}$



Hendra virus survival does not explain spillover patterns and implicates relatively direct transmission routes from flying foxes to horses

James M. Cook¹, James M. Cook¹, James M. Cook¹, James M. Cook¹, James M. Cook¹,
James M. Cook¹ and James M. Cook¹

Correspondence

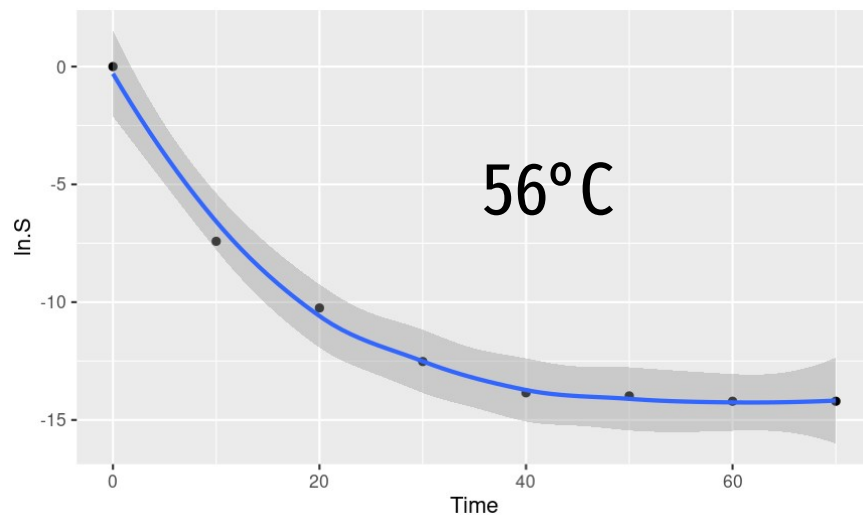
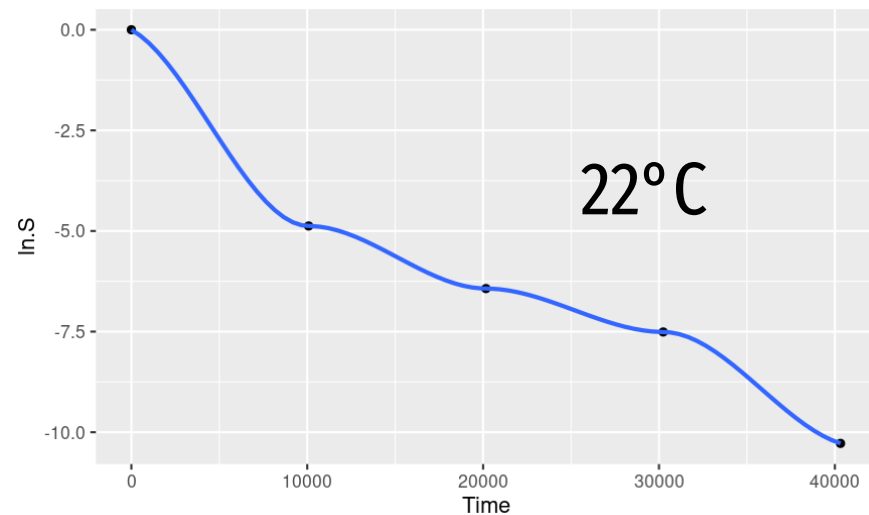
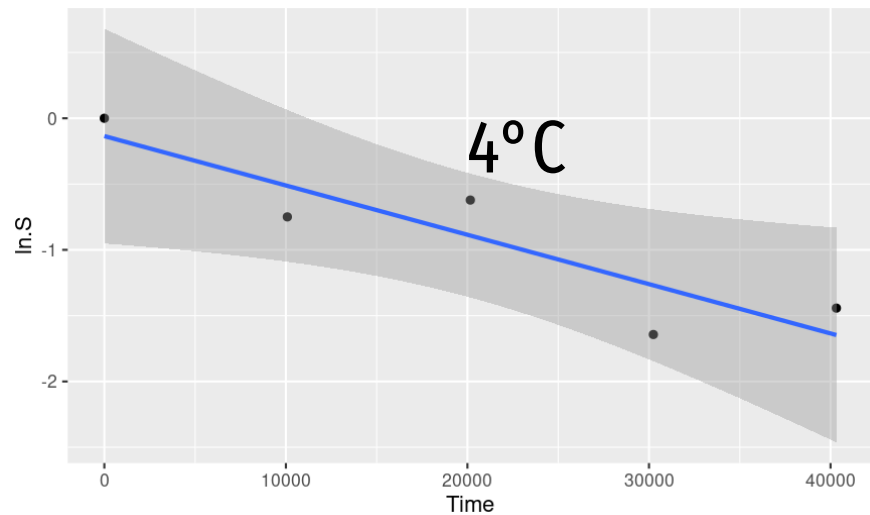
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³Pennsylvania State University, State College, PA, USA

⁴Monash University, Melbourne, Victoria, Australia

⁵Commonwealth Scientific and Industrial Research Organisation, Geelong, Victoria, Australia



Trainer Rail died of same respiratory disease as horses

BRISBANE: Horse trainer Vic Rail died of the same disease which killed 14 horses, and two stablehands had also contracted its symptoms, Queensland Health authorities said yesterday.

Queensland director of Public Health Dr Gerry Murphy and the director of Animal Health, Kevin Dunn, told a joint press conference, that tests on two Rail employees had demonstrated antibodies of the virus but neither were at risk of developing serious illness.

Mr Dunn said authorities were still puzzled how the horses contracted the virus, now being called Acute Equine Respiratory Syndrome, but quarantine measures put in place since the discovery of the virus had played a key role in controlling it.

The Queensland Minister for Primary Industries, Ed Casey, who chaired the conference, also announced a "limited" return to racing at Eagle Farm on Saturday. However the return is only for horses trained within the 5km

quarantine area of Rail's Hendra stables.

Current restrictions on the movements of horses in the area will remain.

Two-year-old barrier trials scheduled for Eagle Farm tomorrow have been postponed again and the Albion Park trot meeting scheduled on Saturday night transferred to the Gold Coast.

The Doomben meeting next Wednesday has already been transferred to Ipswich.

Rail, 49, was well known for his

training feats of former champion racehorse Vo Rogue, who won 26 races and was placed a further 25 times from 83 starts, winning more than \$1.1 million in prize money.

Mr Dunn said tests in the past few days had ruled out a toxic cause in the deaths of the horses in Rail's stable.

"We now know a virus lead to the deaths," Mr Dunn said.

"At this stage it appears the virus is limited to the Hendra premises and it appears it is a virus that is not spread in any highly

contagious way from horse to horse."

"We are not dealing with an epidemic, but at this stage we don't know the extent of the limit of infection of this virus."

"It may be that it is possible to infect horses without causing serious disease."

"This is one of the important areas that is currently under investigation."

Dr Murphy said that the post-mortem on Rail ruled out legionnaire's disease and indicated he

had an illness "very similar, if not identical" to that which killed the horses.

"From tests so far conducted at the Animal Health Laboratory there is strong presumptive evidence that the same virus which killed the horses also killed Mr Rail."

Dr Murphy said evidence also showed the virus was not spread through the air but how it was spread was still unknown.

"It would appear as though very close contact is necessary with in-

fects saliva or blood from a sick horse," he said.

"For that reason and also because there's been no other cases of severe human illness in this present episode leads to a very firm conclusion that there's no risk to the general health of the public."

Mr Dunn said foul play had also been ruled out in Rail's death and police had dropped their investigation.

Health authorities believe the virus may belong to the same virus

group which causes measles in humans, distemper in dogs and a particular virus in cattle.

"We believe it's a single virus but hitherto has been unrecognized as disease causing," Mr Dunn said.

"It's too early to say whether it's an existing virus which may have changed in some way or mutated."

Health authorities hope to receive further test results next week while a further meeting between health authorities and racing officials will be held next Tuesday.

Deadly equine virus in farmer's body: experts

By RORY CALLINAN
and MARK OBERHARDT

MACKAY farmer Mark Preston had equine morbillivirus in his brain fluid before he died, according to an article in a Federal Health Department newsletter.

Mr Preston, 35, died last month after exhibiting symptoms of a brain inflammation-type illness.

Health Minister Peter Beattie's office yesterday said it had the results of the autopsy on Mr Preston but refused to confirm the virus was the cause of his death.

Mr Preston's father said yester-



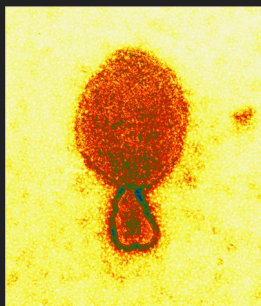
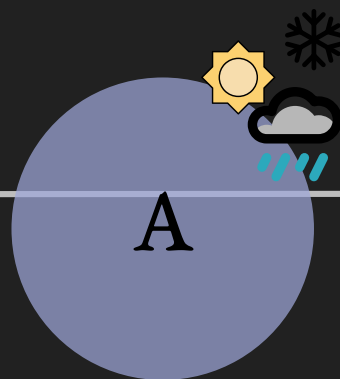
epilepsy before being transferred to the Royal Brisbane Hospital on September 30.

"He became comatose and required ventilation," the article said. "Magnetic resonance imaging showed extensive patchy cerebral disease (widespread brain damage)."

The article noted a difference between the symptoms exhibited by Mr Preston and Mr Rail, who died in September last year.

"(Mr Preston) had no respiratory symptoms until he developed an aspiration pneumonia," it said.

"Major clinical findings in the



Black FF
P. alecto



Grey headed FF
P. poliocephalus

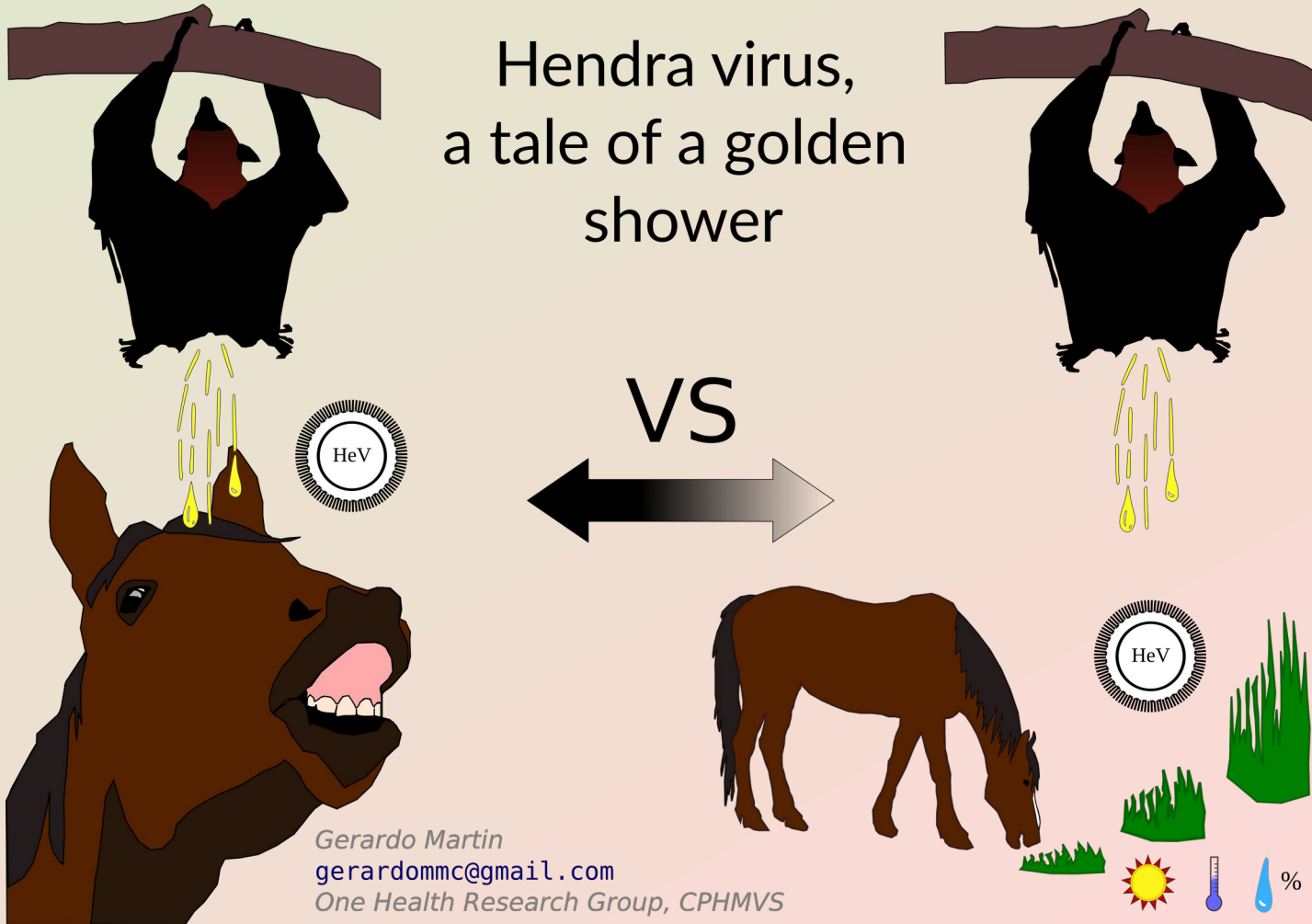


Spectacled FF
P. conspicillatus



Little red FF
P. scapulatus

Hendra virus, a tale of a golden shower



Gerardo Martin
gerardommc@gmail.com
One Health Research Group, CPHMVS

Veremos el resto de esta historia **aquí...**